

# Value Co-Creation Modeling for Mission-Oriented Sustainable Service Ecosystems: An Engineering Management Framework for Campus Food Services

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### 1 Abstract

Mission-oriented service systems pose a fundamental engineering management challenge because value must be created across multiple stakeholders with interdependent objectives, constrained resources, and shared sustainability goals. This paper develops an **engineering management framework** for designing and improving **mission-oriented sustainable service ecosystems** through **value co-creation modeling** and a **Multi-Stakeholder Recommendation System (MSRS)** embedded in a broader decision-support architecture. Rather than treating recommendation as a user-centric personalization tool, the framework conceptualizes MSRS as a coordination mechanism that aligns stakeholder decisions across design, production, delivery, and governance.

The framework is illustrated through the context of **healthy, tasty, and affordable campus food services**, where consumers, producers, operators, and regulators jointly shape service outcomes. The model integrates four elements: stakeholder utility structures, exchange mechanisms, food and service flows, and a digital coordination layer that supports co-design, co-engineering, production planning, delivery execution, and adaptive improvement. In this setting, the MSRS coordinates consumer needs and profiles, producer capabilities and capacities, operator feasibility constraints, and regulatory requirements to improve ecosystem-level performance.

The paper contributes to engineering management in three principal ways. First, it extends value co-creation from a descriptive service concept to a formal design logic for mission-oriented service ecosystems. Second, it reframes multi-stakeholder recommendation systems as

mechanisms for ecosystem coordination rather than only end-user recommendation. Third, it provides a simulation-ready framework for evaluating multidimensional performance outcomes, including service effectiveness, affordability, operational efficiency, sustainability, and innovation. The framework offers a transferable basis for managing sustainable service ecosystems in food and other mission-oriented industries.

**Keywords:** engineering management; mission-oriented innovation; sustainable service ecosystem; value co-creation; multi-stakeholder recommendation system; decision support system; service innovation; campus food services; affordability; sustainability

## 2 Introduction

Mission-oriented service systems pose a distinctive engineering management challenge because value must be created across multiple stakeholders with interdependent objectives, constrained resources, and shared sustainability goals. In such systems, performance depends not only on the quality of a focal product or service, but also on the ability to coordinate actors whose goals are partially aligned yet often conflicting. Consumers seek relevant, affordable, and high-quality outcomes; producers seek viable and scalable offerings; operators seek efficient and reliable execution; and regulators seek compliance with broader public or institutional missions such as sustainability, health, inclusion, and fairness. As a result, managing mission-oriented services requires more than optimizing isolated functions. It requires ecosystem-level coordination across design, production, delivery, and governance.

This challenge is particularly visible in service ecosystems where mission achievement depends simultaneously on market responsiveness and public-value creation. Conventional service design approaches often treat innovation as a sequence led by a focal provider, while conventional recommendation systems typically optimize for end-user relevance alone. Neither perspective is sufficient when sustainable performance depends on coordinated exchanges among heterogeneous stakeholders. In such contexts, value is not produced by one actor and delivered to another; it is co-created through the integration of demand signals, production capabilities, operational constraints, regulatory requirements, and continuous feedback.

This paper addresses that problem by proposing a **value co-creation modeling framework** for **mission-oriented sustainable service ecosystems** supported by a **Multi-Stakeholder Recommendation System (MSRS)**. The core argument is that MSRS should be understood not merely as a recommendation tool, but as a digital coordination mechanism that helps align stakeholder decisions with mission-level objectives. The framework focuses on four stakeholder groups—**consumers, producers, operators, and regulators**—and explains how their interactions can be structured through exchange mechanisms that support co-design, co-engineering, production, delivery, consumption, and continuous reconfiguration.

The framework is illustrated through the case of **healthy, tasty, and affordable campus food services**, which provides a bounded yet representative setting for examining mission-

oriented service design. This context is particularly suitable because campus food services combine product and service design, operational execution, sustainability pressures, affordability constraints, and institutional mission goals. In this setting, consumers shape demand through preferences and profiles, producers shape feasible offerings through capabilities and innovation options, operators shape service performance through capacity and delivery decisions, and regulators shape mission alignment through standards, incentives, and governance. The proposed MSRS coordinates these interactions by balancing stakeholder utilities rather than privileging a single actor’s objective.

The paper contributes to engineering management in three principal ways. First, it develops a mission-oriented view of value co-creation in which stakeholder alignment is treated as a design problem rather than an emergent by-product of service exchange. Second, it reframes MSRS as a mechanism for ecosystem orchestration that connects stakeholder utilities to mission performance across the full service lifecycle. Third, it offers a simulation-ready framework for evaluating ecosystem performance in terms of effectiveness, affordability, operational efficiency, sustainability, and innovation. By doing so, the paper provides an engineering management perspective on how sustainable service ecosystems can be designed, governed, and improved through coordinated digital decision support.

The remainder of the paper is organized as follows. Section 2 develops the theoretical framework and propositions. Section 3 explains the process model of co-design and co-engineering in mission-oriented service ecosystems. Section 4 presents the system architecture and decision-support logic. Section 5 outlines the multi-agent and simulation-based evaluation design. Section 6 reports illustrative results. Section 7 discusses theoretical and managerial implications. The final section concludes and identifies directions for future research.

## 3 Literature Review Design

### 3.1 Review objective

The review aimed to identify and synthesize research relevant to a **healthy-food ecosystem for university campuses using multi-stakeholder recommendation systems**. Because this exact topic is still emerging, the review used an interdisciplinary search strategy spanning four linked domains:

1. campus food environments,
2. food and healthy-food recommender systems,
3. multi-stakeholder recommendation and fairness,
4. sustainability and food-service operations.

### 3.2 Search strategy

A rapid systematic review logic was applied using targeted scholarly web searching across the four domains above. Search terms included combinations of the following concepts:

- multi-stakeholder recommendation,
- food recommender systems,
- healthy food recommendation,
- campus food environment,
- university food environment,
- university restaurants sustainability,
- food waste in food services.

Priority was given to review papers, foundational conceptual papers, technical studies with clear relevance, and implementation-oriented studies.

### 3.3 Inclusion criteria

Studies were included if they contributed directly to at least one of the following:

1. university or campus food environments,
2. food or healthy-food recommender systems,
3. multi-stakeholder recommendation, multi-sided fairness, or multi-objective recommendation,
4. sustainability, food-service operations, or waste reduction in institutional dining.

### 3.4 Exclusion criteria

Studies were excluded if they:

- were unrelated to food, recommendation, or institutional dining,
- were too general to inform the proposed framework,
- or did not contribute meaningful conceptual, empirical, or methodological insight.

### 3.5 Screening and selection logic

Screening followed a staged logic. First, potentially relevant studies were identified across the four domains. Second, titles and abstracts or result snippets were screened for conceptual relevance. Third, full-text or source-level pages were assessed for eligibility. Studies were retained if they offered direct evidence on campus food behavior or institutional food systems, technical or conceptual foundations for healthy-food recommendation, theoretical grounding

for multi-stakeholder recommendation, or sustainability and operational insights relevant to food-service coordination.

### **3.6 Study Selection Procedure**

This review followed a rapid systematic literature review approach across four related domains: campus food environments, food and healthy-food recommender systems, multi-stakeholder recommendation and fairness, and sustainability or food-service operations. Because the topic remains emergent and interdisciplinary, the review used a PRISMA-style flow logic rather than a full database-exhaustive PRISMA 2020 procedure.

healthy\_food\_ecosystem\_slr\_paper\_files/figure-latex/mermaid-figure-1.png

**Figure 1. PRISMA-style flow of study selection.** Records were identified through targeted scholarly searching across four domains: campus food environments, food recommender systems, multi-stakeholder recommendation, and sustainability or food-service operations. After screening and eligibility assessment, 20 key studies were included in the final thematic synthesis.

## 4 Theoretical Background

### 4.1 Campus food environment research

The campus food environment literature consistently shows that university food settings shape dietary behavior through availability, affordability, convenience, social context, and environmental design (Roy et al. 2015; Deliens et al. 2014; Caruso et al. 2025). This stream establishes the university as a legitimate unit of intervention and supports the view that food choice is shaped by institutional conditions rather than personal preference alone.

### 4.2 Food and healthy-food recommender systems

The food recommender systems literature indicates that food recommendation is now a well-developed subfield with a distinct methodological identity (Bondevik et al. 2024; Tran et al. 2018; Trattner and Elswiler 2017). Common methods include content-based recommendation, collaborative filtering, hybrid models, and machine-learning approaches. However, most systems remain strongly user-centric even when they include health-aware logic.

### 4.3 Multi-stakeholder recommendation and fairness

The multi-stakeholder recommendation literature provides the strongest theoretical foundation for the proposed framework because it explicitly addresses the coordination of multiple legitimate utilities within a shared decision space (Zheng 2017; Abdollahpouri and Burke 2019; Wu et al. 2021). This stream argues that recommendation environments often involve several affected parties and that recommendation quality should therefore be evaluated across multiple utilities rather than only end-user satisfaction.

### 4.4 Sustainability and food-service operations

The sustainability and institutional food-service literature emphasizes food waste, restaurant sustainability, and operational coordination, thereby highlighting the need to connect environmental performance with service design and execution (Peixoto et al. 2026; Guimaraes et al. 2024). This stream is especially relevant for canteen planning, procurement, and waste

reduction, although it is still only weakly connected to healthy-food recommender system design.

## 5 Review Synthesis and Research Gap

The reviewed literature can be synthesized into four interrelated streams: **campus food environments**, **food and healthy-food recommender systems**, **multi-stakeholder recommendation**, and **sustainability and food-service operations**. Taken together, these streams strongly support the relevance of the proposed topic, but they also show that the field remains fragmented and that an integrated university-campus framework is still underdeveloped (Bondevik et al. 2024; Caruso et al. 2025; Zheng 2017; Peixoto et al. 2026).

### 5.1 1. Campus food environments as intervention settings

A consistent finding across campus food studies is that university food environments shape dietary behavior through factors such as availability, affordability, convenience, time pressure, and social context (Roy et al. 2015; Deliens et al. 2014; Caruso et al. 2025). This means that healthy eating on campus cannot be reduced to individual preference alone. Any recommendation framework intended for university campuses must therefore be embedded in the actual food environment rather than designed as an isolated personalization tool.

### 5.2 2. Food recommender systems are mature, but mainly user-centric

The food recommender systems literature is now sufficiently mature to support domain-specific design choices (Bondevik et al. 2024; Tran et al. 2018; Trattner and Elswiler 2017). However, most existing systems remain centered on the individual user. Even when they incorporate health-aware logic, they usually optimize for some combination of user preference, dietary fit, and nutritional value, while paying limited attention to food-service feasibility, supplier constraints, institutional objectives, or environmental outcomes.

### 5.3 3. Multi-stakeholder recommendation offers the strongest theoretical foundation

The multi-stakeholder recommendation literature provides the clearest conceptual basis for the proposed healthy-food ecosystem (Zheng 2017; Abdollahpouri and Burke 2019; Wu et al. 2021). This literature is highly relevant because campus food systems are inherently multi-actor environments. Consumers seek taste, convenience, affordability, and health; canteens seek manageable production and waste reduction; suppliers seek predictable demand; and administrators may prioritize sustainability and wellbeing.



#### **5.4 4. Sustainability and food-service research remain insufficiently integrated with recommendation**

The sustainability and food-service literature adds an important operational perspective (Peixoto et al. 2026; Guimaraes et al. 2024). Research in this stream still focuses heavily on waste reduction, especially post-distribution and plate waste. While this is important, the literature provides weaker integration between sustainability metrics, nutritional goals, and recommendation-system design.

#### **5.5 5. Explainability, trust, and real-world deployment are emerging priorities**

Newer implementation-oriented studies indicate that healthy-food recommendation is moving beyond technical prototypes toward real-life settings (Authors 2025; Wang et al. 2024; Rostami et al. 2024). This has direct implications for campus adoption. A university healthy-food ecosystem would likely require not only accurate recommendation, but also interpretable suggestions, stakeholder buy-in, and policy legitimacy.

#### **5.6 6. Overall synthesis and gap**

Across the reviewed streams, the literature supports three major conclusions (Bondevik et al. 2024; Caruso et al. 2025; Zheng 2017; Peixoto et al. 2026). First, the campus is a valid and influential food environment. Second, healthy-food recommendation is technically feasible and increasingly sophisticated. Third, multi-stakeholder recommendation provides an appropriate framework for balancing competing interests. What remains missing is an integrated model that combines these insights into a single university-campus food ecosystem connecting consumers, canteens, providers, suppliers, researchers, and administrators.

#### **5.7 Evidence Base Overview: 20 Key Papers**

| # | Paper                                                                                                                            | Stream          | Aim                                    | Method                  | Main finding                                                            | Why it matters                              |
|---|----------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------|-------------------------|-------------------------------------------------------------------------|---------------------------------------------|
| 1 | Bondevik et al. (2024), <i>A systematic review on food recommender systems</i>                                                   | Food RS         | Review the food recommender field      | Systematic review       | Food recommenders are mature but fragmented across methods and goals    | High-level overview of food RS              |
| 2 | Tran et al. (2018), <i>An overview of recommender systems in the healthy food domain</i>                                         | Healthy food RS | Survey healthy-food recommenders       | Technical review        | Healthy-food RS must balance preference, nutrition, and behavior change | Foundation for healthy recommendation logic |
| 3 | Trattner & Elsweiler (2017), <i>Food Recommender Systems: Important Contributions, Challenges and Future Research Directions</i> | Food RS         | Summarize contributions and challenges | Scholarly survey        | Food RS differ from generic RS due to health and context                | Seminal conceptual framing                  |
| 4 | Ge et al. (2015), <i>Health-aware Food Recommender System</i>                                                                    | Healthy food RS | Introduce a health-aware recommender   | Conference system paper | Health can be embedded in recommendation logic                          | Early nutrition-aware system                |

| # | Paper                                                                                                           | Stream                 | Aim                                          | Method                         | Main finding                                                  | Why it matters                      |
|---|-----------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------|--------------------------------|---------------------------------------------------------------|-------------------------------------|
| 5 | Rostami et al. (2024), <i>A novel healthy food recommendation to user groups...</i>                             | Healthy food RS        | Move from individual to group recommendation | Model + experiments            | Group recommendation is feasible in food contexts             | Relevant for shared dining settings |
| 6 | Wang et al. (2024), <i>Improving healthy food recommender systems through heterogeneous hypergraph learning</i> | Healthy food RS        | Improve recommendation accuracy              | ML model                       | Richer user-food-ingredient modeling improves personalization | Useful for technical engine design  |
| 7 | Workplace explanation study (2025)                                                                              | Explainability         | Study explanations in deployment             | User-centered deployment study | Explanations improve understanding and control                | Relevant for trust and adoption     |
| 8 | Tsolakidis et al. (2024), <i>AI and ML Technologies for Personalized Nutrition: A Review</i>                    | Personalized nutrition | Review AI/ML for nutrition                   | Review                         | Personalized nutrition is increasingly recommender-driven     | Bridge to nutrition science         |

| #  | Paper                                                                                                           | Stream            | Aim                                   | Method                    | Main finding                                          | Why it matters                       |
|----|-----------------------------------------------------------------------------------------------------------------|-------------------|---------------------------------------|---------------------------|-------------------------------------------------------|--------------------------------------|
| 9  | Zheng (2017), <i>Multi-Stakeholder Recommendation: Applications and Challenges</i>                              | MSR               | Define MSR and agenda                 | Conceptual paper          | Recommendation involves multiple legitimate utilities | Core theory                          |
| 10 | Abdollahpour & Burke (2019), <i>Multi-stakeholder Recommendation and its Connection to Multi-sided Fairness</i> | MSR               | Connect MSR with fairness             | Conceptual paper          | Fairness is central in multi-stakeholder settings     | Governance foundation                |
| 11 | Wu et al. (2021), <i>Multi-FR</i>                                                                               | Fairness-aware RS | Optimize across stakeholder interests | Multi-objective framework | Multi-objective optimization can balance utilities    | Strong precedent for scoring models  |
| 12 | Burke et al. (2018), <i>Balanced neighborhoods for multi-sided fairness in recommendation</i>                   | Fairness-aware RS | Improve fairness in exposure          | Fairness-aware method     | Rankings can reduce one-sided bias                    | Relevant for equitable food exposure |

| #  | Paper                                                                                    | Stream      | Aim                                              | Method                  | Main finding                                              | Why it matters                |
|----|------------------------------------------------------------------------------------------|-------------|--------------------------------------------------|-------------------------|-----------------------------------------------------------|-------------------------------|
| 13 | Tran et al. (2021), <i>Recommender systems in the healthcare domain</i>                  | Health RS   | Review healthcare RS                             | State-of-the-art review | Food RS fit within digital health RS                      | Useful positioning paper      |
| 14 | Roy et al. (2015), <i>Food environment interventions in tertiary education settings</i>  | Campus food | Review interventions                             | Systematic review       | Tertiary food interventions can improve choices           | Strong campus evidence        |
| 15 | Deliens et al. (2014), <i>Determinants of eating behaviour in university students</i>    | Campus food | Identify drivers of eating behavior              | Qualitative study       | Taste, convenience, peers, price, and availability matter | Supports context-aware design |
| 16 | Caruso et al. (2025), <i>The campus food environment and post-secondary student diet</i> | Campus food | Review links between campus environment and diet | Systematic review       | Campus food environments influence diet                   | Supports ecosystem framing    |

| #  | Paper                                                                                  | Stream                    | Aim                                      | Method                            | Main finding                                               | Why it matters                        |
|----|----------------------------------------------------------------------------------------|---------------------------|------------------------------------------|-----------------------------------|------------------------------------------------------------|---------------------------------------|
| 17 | Dahl et al. (2024), <i>Assessing the Healthfulness of University Food Environments</i> | Campus food / measurement | Review methods and tools                 | Systematic review                 | Assessment methods are heterogeneous                       | Useful for KPI design                 |
| 18 | Peixoto et al. (2026), <i>Sustainability in university restaurants</i>                 | Sustainability            | Map strategies in university restaurants | Scoping review                    | Research is still waste-centric                            | Reveals gap for integration           |
| 19 | Guimarães et al. (2024), <i>Plate Food Waste in Food Services</i>                      | Operations / waste        | Quantify plate waste                     | Systematic review + meta-analysis | Food waste is measurable and significant                   | Useful for canteen objectives         |
| 20 | <i>Recommender systems and sustainability: a dual perspective</i> (2026)               | Sustainability + RS       | Review sustainability in RS              | Literature review                 | Sustainability should be both outcome and design principle | Relevant for environmental objectives |

## 5.8 Research Gap Summary

The review identifies five major gaps:

1. **No integrated campus framework** combining consumers, canteens, providers, suppliers, researchers, and administrators in one recommendation ecosystem.
2. **Operational constraints** such as kitchen capacity, inventory, and procurement are rarely integrated into healthy-food recommendation models.
3. **Sustainability variables** such as sourcing, seasonality, and emissions remain weakly linked to recommendation objectives.

4. **Real-world institutional deployments** are still limited.
5. **Governance issues** such as fairness, explainability, consent, and institutional priority weighting are underdeveloped in food-specific applications.

## 6 Theoretical Framework and Propositions

### 6.1 Theoretical framing

Mission-oriented services differ from conventional service systems because they are designed to achieve not only market or organizational performance, but also broader societal and institutional goals such as sustainability, health, inclusion, resilience, and wellbeing. In such systems, value cannot be adequately explained through dyadic provider-consumer exchange alone. Instead, value emerges through interactions among multiple actors whose interests are distinct, partially aligned, and interdependent. This requires a broader theoretical perspective that combines **value co-creation**, **service ecosystems**, and **multi-stakeholder coordination**.

The framework proposed in this study is grounded in three complementary ideas. First, value co-creation theory suggests that value is created through interaction, resource integration, and mutual adjustment among actors rather than transferred unilaterally from producer to user. Second, service ecosystem theory emphasizes that value creation occurs within institutionalized structures of actors, rules, technologies, and resource flows. Third, multi-stakeholder recommendation logic provides a mechanism for operationalizing coordination across heterogeneous actors by explicitly modeling their utilities and balancing them within a shared decision framework.

Bringing these perspectives together, this study argues that mission-oriented services can be understood as **sustainable value co-creation ecosystems** in which consumers, producers, regulators, and operators jointly contribute to both stakeholder-specific and mission-level outcomes. Within this ecosystem, the **Multi-Stakeholder Recommendation System (MSRS)** acts as the coordinating intelligence that aligns decentralized choices with broader mission goals.

### 6.2 Core constructs

#### 6.2.1 Mission-oriented service ecosystem

A mission-oriented service ecosystem is defined here as a structured network of interacting actors, resources, rules, and technologies organized around a shared public or strategic objective. Unlike ordinary service systems, mission-oriented ecosystems must balance private interests with public outcomes.

### 6.2.2 Stakeholder value co-creation

Stakeholder value co-creation refers to the process by which consumers, producers, regulators, and operators jointly generate value through interdependent exchanges of information, resources, constraints, incentives, and feedback. In this view, value is not created by one actor and consumed by another; it is co-produced through coordination.

### 6.2.3 Exchange mechanism

The exchange mechanism is the set of structured flows through which stakeholders interact. These flows include demand exchange, supply exchange, operational exchange, regulatory exchange, and feedback exchange. These exchanges are the channels through which co-creation becomes actionable.

### 6.2.4 Multi-Stakeholder Recommendation System

The MSRS is conceptualized as a decision-orchestration mechanism that evaluates alternatives across multiple stakeholder utilities simultaneously. It generates differentiated recommendations for different actor groups and therefore supports mission alignment, not only personalization.

### 6.2.5 Mission performance

Mission performance refers to the degree to which the ecosystem achieves shared societal or institutional objectives, such as sustainability, health, affordability, fairness, or resilience, while maintaining stakeholder viability.

## 6.3 Propositions

**Proposition 1.** The effectiveness of mission-oriented services increases when value creation is modeled as a multi-stakeholder co-creation process rather than as a dyadic provider-consumer exchange.

**Proposition 2.** Structured exchange mechanisms among consumers, producers, regulators, and operators positively influence value co-creation in mission-oriented service ecosystems.

**Proposition 3.** The greater the alignment between stakeholder-specific objectives and mission-level goals, the higher the overall performance of the service ecosystem.

**Proposition 4.** An MSRS positively affects mission-oriented ecosystem performance by improving coordination across stakeholder exchanges.



**Proposition 5.** Compared with user-centric recommendation systems, MSRS-based coordination leads to superior trade-offs among quality, affordability, efficiency, and sustainability.

**Proposition 6.** The contribution of an MSRS to ecosystem performance is strengthened when governance mechanisms such as fairness rules, explainability, and mission-weighting are explicitly embedded in the system.

**Proposition 7.** Mission-oriented service ecosystems supported by MSRS are more likely to generate sustainable innovation outcomes than ecosystems coordinated through static or rule-based mechanisms alone.

## 6.4 Theoretical contribution

The proposed theoretical framing contributes to engineering management by extending value co-creation from a largely dyadic service logic to a mission-oriented ecosystem perspective, by repositioning MSRS as a mechanism for ecosystem-level coordination rather than only consumer personalization, and by linking mission-oriented innovation with digitally enabled governance of service performance.

# 7 Engineering Management Framework

## 7.1 Core proposition

Based on the literature and the theoretical framing developed above, this paper proposes that mission-oriented service ecosystems should be designed around a **multi-stakeholder coordination architecture** rather than around isolated consumer-facing tools or function-specific optimizations. In the campus food-service application, this means shifting from a consumer-only recommendation logic to a broader architecture in which recommendation operates as an ecosystem-level coordination mechanism across multiple actors.

## 7.2 Key stakeholders

The proposed ecosystem includes:

- consumers: students, professors, support staff,
- canteens and food-service operators,
- food providers and caterers,
- raw-food suppliers,
- healthy-food researchers,
- university administration.

### 7.3 Core system modules

A future implementation should include:

- personalized meal recommendation,
- group or shared-choice recommendation,
- multi-objective optimization,
- demand forecasting,
- menu optimization,
- procurement recommendation,
- sustainability scoring,
- explanation interfaces,
- intervention and nudge testing,
- governance mechanisms for fairness and consent.

### 7.4 Process Model for Co-Design and Co-Engineering in Mission-Oriented Service Ecosystems

The proposed ecosystem can be understood as a **closed-loop mission-oriented innovation process** in which stakeholders jointly shape the design, production, delivery, use, and continuous improvement of products or services. Rather than treating innovation as a linear process controlled by a single provider, the framework conceptualizes it as a multi-stakeholder cycle in which consumers, producers, operators, and regulators continuously exchange information, constraints, capabilities, and feedback. Within this process, the **Multi-Stakeholder Recommendation System (MSRS)** acts as the coordination mechanism that aligns stakeholder decisions with both individual interests and mission-level goals.

The process begins with a **sensing stage**, in which the ecosystem collects information about consumer needs, usage contexts, price sensitivity, quality expectations, and profiles, alongside producer capabilities, operator capacities, and regulatory requirements. At this stage, the purpose is not yet to produce a final offering, but to construct a shared opportunity space in which innovation problems and possibilities can be identified. The MSRS aggregates and interprets these heterogeneous signals to identify promising directions for ecosystem-level action.

The second stage is **co-design**, in which candidate products or services are defined collaboratively. In this stage, consumers contribute information about desirable attributes, producers contribute design alternatives and innovation possibilities, operators identify delivery and execution constraints, and regulators ensure that concepts remain aligned with standards and mission objectives. The output of this stage is not a single final offering, but a portfolio of feasible concepts that can be evaluated against stakeholder utilities. The MSRS contributes by ranking these alternatives according to expected consumer relevance, producer feasibility, operator practicality, and regulatory alignment.

The third stage is **co-engineering**, in which promising concepts are transformed into feasible product-service configurations. Here, the ecosystem moves from “what should be offered” to “how it should be produced and delivered.” Producers determine feasible production structures, operators design service and delivery workflows, regulators verify compliance and performance standards, and consumers validate expected usability and convenience. In this stage, the MSRS functions as an engineering coordination tool by identifying the combination of product attributes, resource choices, operational settings, and service conditions that best satisfies ecosystem-level objectives.

The fourth stage is **production and resource allocation**, in which the selected offering is prepared for execution. This includes sourcing materials, assigning production capacity, scheduling operations, and balancing cost and quality constraints. Although producers and operators play the most visible roles in this stage, consumer demand signals, regulatory constraints, and MSRS-guided forecasts continue to shape decisions. The purpose of this stage is to ensure that the offering is not only desirable in principle, but also producible within available capabilities and constraints.

The fifth stage is **delivery and service execution**, in which the offering is made accessible to consumers in a timely, convenient, and reliable manner. This stage is especially important because mission-oriented value can fail even when design quality is high if service execution is poor. Operators are central here, but producers support quality continuity, regulators monitor service standards, and consumers directly experience convenience and timeliness. The MSRS contributes by coordinating timing, availability, and service channels so that the final delivery configuration remains aligned with stakeholder conditions.

The sixth stage is **consumption and value realization**, where the offering is actually used, experienced, and evaluated. At this point, value is no longer only potential; it becomes realized through actual usage. Consumers generate satisfaction, dissatisfaction, adoption, or rejection; producers observe acceptance and innovation success; operators observe service outcomes; and regulators observe mission-related impacts. The ecosystem therefore treats consumption not as the end of the process, but as a key information-generating stage.

The seventh stage is **evaluation**, in which stakeholder outcomes and mission outcomes are assessed together. Performance in this stage is measured not only in terms of user satisfaction, but also in terms of affordability, operational efficiency, compliance, sustainability, and innovation success. The MSRS integrates these outcomes into a multi-stakeholder performance profile that reveals both synergies and trade-offs. This enables the ecosystem to evaluate whether value has been jointly created or whether one stakeholder’s gain has come at the expense of another.

The final stage is **reconfiguration and improvement**, in which the ecosystem adapts based on observed results. Offerings may be redesigned, production processes improved, delivery models adjusted, incentives recalibrated, or recommendation logic updated. In this sense, the ecosystem is not static; it is a continuously learning and co-evolving system. The MSRS

supports this adaptive process by updating its coordination logic according to feedback from previous cycles.

Taken together, this process model shows that mission-oriented innovation in service ecosystems is fundamentally **iterative, multi-actor, and feedback-driven**. Consumers shape demand and realized value, producers shape feasible offerings, operators shape execution quality, and regulators shape mission compliance and public legitimacy. The MSRS links these actors across the full lifecycle—from sensing and design to production, delivery, consumption, evaluation, and redesign—thereby enabling sustainable value co-creation rather than isolated optimization.

## 8 Evaluation Design

### 8.1 Methodology

#### 8.1.1 Research design

This study adopts a **design science and simulation-based research** approach to develop and evaluate a **multi-stakeholder healthy-food ecosystem** for university campuses. The proposed artifact is an integrated socio-technical system composed of four tightly connected layers: **stakeholders, food flow, decision support**, and a **multi-stakeholder recommendation system (MSRS)**. For experimental evaluation, the ecosystem is specified as a **multi-agent system (MAS)** in which autonomous stakeholders interact within a shared campus food-service environment. Formally, the system may be represented as  $MAS = (A, E, F, O, P, U, T)$ , where  $A$  denotes the set of agents,  $E$  the environment,  $F$  the food-flow states and transitions,  $O$  the observations available to each agent,  $P$  the decision policies,  $U$  the stakeholder-specific utility functions, and  $T$  the temporal transition rules. The agent set includes consumers, canteens, suppliers, research/policy actors, administrators, and an MSRS coordinator. The environment contains menus, inventory, kitchen capacity, supplier pipelines, prices, queues, waste, and policy settings, while food entities move through state-based transitions from raw ingredients to preparation, serving, consumption, and waste. Each agent observes a subset of the system state and chooses actions according to its objectives and constraints. Consumer agents maximize taste, health, affordability, and convenience; canteen agents maximize margin, feasibility, and service quality while minimizing waste; supplier agents maximize order stability and fulfillment performance; and administrative actors maximize health, sustainability, fairness, and affordability outcomes. The MSRS coordinator agent ranks candidate actions using a weighted ecosystem objective and generates coordinated recommendations for meal choice, menu planning, procurement, pricing, and interventions. This framing is appropriate because multi-stakeholder recommendation research treats recommendation as a setting with multiple legitimate utilities, while agrifood digital-twin and simulation literature supports virtual experimentation for food-flow, operational, and sustainability decisions (Zheng 2017; Abdollahpouri and Burke 2019; Wu et al. 2021).

### 8.1.2 Ecosystem architecture

The ecosystem is modeled as a layered architecture:

1. **Stakeholder layer**
2. **Food-flow layer**
3. **Decision-support layer**
4. **MSRS optimization layer**
5. **Governance and monitoring layer**

This layered structure integrates physical food operations with digital intelligence and coordinated decision-making, thereby linking ecosystem design with operational execution (Zheng 2017; Wu et al. 2021).

### 8.1.3 Stakeholder model

The stakeholder layer encompasses the following actor groups:

- **Consumers:** students, faculty, and staff
- **Food-service operators:** canteens, caterers, kitchen managers
- **Supply actors:** raw-food suppliers, processors, distributors
- **Governance actors:** nutrition researchers, sustainability office, university administration

Each stakeholder group  $s$  is represented by: - a utility function  $U_s$ , - constraints  $C_s$ , - decisions  $D_s$ , - and performance measures  $K_s$ .

Typical utility structures include: - consumer utility: taste, affordability, health, convenience; - canteen utility: margin, feasibility, throughput, waste reduction; - supplier utility: demand stability, order visibility, spoilage reduction; - university utility: health promotion, sustainability, fairness, and policy compliance.

### 8.1.4 Food-flow model

The food system is represented as a directed network:

$$G = (N, E)$$

where  $N$  denotes process nodes and  $E$  denotes flow links. The main food-flow chain is:

**Raw material supply** → **inbound logistics** → **storage** → **preprocessing** → **recipe assembly** → **cooking/preparation** → **serving/display** → **purchase** → **consumption** → **leftovers/returns** → **waste or recovery**

Each food entity is characterized by: - quantity, - unit cost, - lead time, - shelf life, - nutritional profile, - expected taste score, - sustainability score or carbon footprint, - preparation time, - spoilage/waste risk, - selling price.

### **8.1.5 Decision-support system**

The DSS is organized into four interdependent modules.

#### **8.1.5.1 Data module**

- POS transactions
- recipe and nutrition databases
- inventory and procurement records
- supplier deliveries
- queue and service-time data
- user preferences and ratings
- waste audits
- sustainability attributes

#### **8.1.5.2 Analytics module**

- demand forecasting
- inventory monitoring
- waste prediction
- taste preference learning
- price sensitivity estimation
- nutrition scoring
- sustainability scoring

#### **8.1.5.3 Decision module**

- menu planning
- procurement planning
- production scheduling
- dynamic pricing and subsidy suggestions
- recommendation ranking
- intervention and nudge selection

#### 8.1.5.4 Execution module

- consumer app
- canteen dashboard
- supplier portal
- researcher dashboard
- administrative control panel

#### 8.1.6 MSRS model

The MSRS evaluates candidate actions  $a$  through a weighted multi-stakeholder objective:

$$Score(a) = \sum_s w_s U_s(a)$$

where: -  $U_s(a)$  is the utility of stakeholder  $s$  under action  $a$ , -  $w_s$  is the policy weight assigned to stakeholder  $s$ .

The MSRS generates differentiated outputs for multiple actors: - **Consumers:** meal recommendations, bundles, substitutions - **Canteens:** prep quantities, menu mix adjustments, markdown timing - **Suppliers:** replenishment plans, sourcing alternatives - **Researchers/administrators:** nudges, labels, subsidy rules, intervention placement

#### 8.1.7 Governance layer

A cross-cutting governance layer oversees the entire architecture through: - fairness controls, - explainability, - consent and privacy management, - policy-weight management, - KPI monitoring and auditability.

#### 8.1.8 Simulation strategy

To evaluate the proposed ecosystem, the study adopts a **hybrid simulation design**:

- **Agent-based modeling (ABM)** for stakeholder behavior and adaptation
- **Discrete-event simulation (DES)** for kitchen operations, queues, inventory depletion, and deliveries
- **System dynamics (SD)** for longer-term feedback loops such as demand-waste interactions and policy effects

### 8.1.9 Experimental scenarios

The evaluation compares four scenarios:

1. **No recommender**
2. **Consumer-only recommender**
3. **Rule-based healthy menu planning**
4. **Full MSRS-enabled ecosystem**

Experimental factors include: - demand volatility, - budget pressure, - supplier disruption, - taste-preference heterogeneity, - subsidy intensity, - sustainability-priority settings.

### 8.1.10 Evaluation metrics

Performance is assessed across six dimensions:

- **Consumer:** healthy meal uptake, taste satisfaction, affordability compliance, recommendation acceptance
- **Operations:** forecast accuracy, wait time, stockouts, waste, capacity use
- **Financial:** margin, meal cost, subsidy efficiency
- **Sustainability:** carbon footprint per meal, local/seasonal sourcing, total waste
- **Supplier:** order variability, fill rate, spoilage
- **Governance:** fair access, utility balance, explainability/trust

Improvement should be demonstrated through: 1. **statistical significance** across simulation replications, 2. **practical significance** using relative gains and effect sizes, 3. **Pareto improvement** across health, taste, price, and sustainability trade-offs.

## 8.2 System Architecture Description

The proposed architecture links the physical flow of food with digital recommendation and governance.

### 8.2.1 Stakeholder layer

The top layer contains all actors: consumers, canteens, suppliers, researchers, and administrators. These actors contribute data, receive recommendations, and adapt their decisions over time.



### 8.2.2 Food-flow and operations layer

This layer represents the physical transformation chain from raw ingredients to ready-to-consume meals and waste streams:

**Raw ingredients** → **logistics** → **storage** → **preprocessing** → **assembly** → **cooking** → **serving** → **purchase** → **consumption** → **leftovers/waste**

### 8.2.3 Data integration layer

This layer aggregates: - user profiles, - transactions, - recipes and nutrition, - stock and procurement, - delivery records, - queue data, - waste logs, - sustainability indicators.

### 8.2.4 Analytics and DSS layer

This layer transforms data into decisions through: - forecasting, - waste analytics, - nutrition and taste modeling, - price modeling, - sustainability scoring, - intervention analytics.

### 8.2.5 MSRS optimization layer

This is the coordination core. It ranks meals and actions according to multi-stakeholder utility and returns tailored outputs to each actor group.

### 8.2.6 Governance and monitoring layer

This layer overlays all others and manages: - fairness, - transparency, - privacy, - policy priorities, - KPI reporting.

### 8.2.7 Value co-creation logic

Value is co-created when coordinated recommendations improve several stakeholder objectives simultaneously: - **product innovation** through new recipes and substitutions, - **effectiveness** through healthier and more accepted choices, - **efficiency** through lower waste and better forecasting, - **taste** through improved preference-fit, - **price** through affordability and sourcing efficiency.

## 8.3 Simulation Experiment Design

Table 1: Simulation experiment design for evaluating the MSRS-enabled campus healthy-food ecosystem.

| Component                    | Specification                                                                                                                                             | Rationale                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>Objective</b>             | Evaluate whether an MSRS-enabled campus food ecosystem improves health, taste, affordability, efficiency, and sustainability relative to baseline systems | Tests ecosystem-level value of MSRS                                       |
| <b>Model type</b>            | Hybrid <b>ABM + DES + SD</b>                                                                                                                              | Captures heterogeneous actors, operational events, and long-term feedback |
| <b>Unit of analysis</b>      | Campus food ecosystem over daily, weekly, and semester horizons                                                                                           | Supports both operational and strategic evaluation                        |
| <b>Agents</b>                | Consumers, canteens, suppliers, researchers, administrators                                                                                               | Represents multi-stakeholder behavior                                     |
| <b>Food entities</b>         | Ingredients, recipes, prepared meals, served meals, leftovers, waste streams                                                                              | Represents end-to-end food flow                                           |
| <b>Decision engine</b>       | Forecasting + DSS + MSRS optimizer                                                                                                                        | Simulates coordinated decision support                                    |
| <b>Baseline scenario 1</b>   | No recommender                                                                                                                                            | Pure operational reference                                                |
| <b>Baseline scenario 2</b>   | Consumer-only recommender                                                                                                                                 | Compares against standard personalization                                 |
| <b>Baseline scenario 3</b>   | Rule-based healthy menu planning                                                                                                                          | Compares against non-learning policy approach                             |
| <b>Treatment scenario</b>    | Full MSRS-enabled ecosystem                                                                                                                               | Main intervention condition                                               |
| <b>Time step</b>             | Daily operational step; weekly planning cycle; semester evaluation                                                                                        | Matches food-service decision frequency                                   |
| <b>Demand conditions</b>     | Low, medium, high volatility                                                                                                                              | Tests robustness to uncertainty                                           |
| <b>Budget conditions</b>     | Low, medium, high consumer budget pressure                                                                                                                | Tests affordability sensitivity                                           |
| <b>Supply conditions</b>     | Stable supply vs disrupted supply                                                                                                                         | Tests resilience                                                          |
| <b>Preference conditions</b> | Homogeneous vs heterogeneous taste-health preferences                                                                                                     | Tests personalization value                                               |

| Component                      | Specification                                                                                                 | Rationale                               |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Policy conditions</b>       | Different subsidy levels and sustainability weights                                                           | Tests policy sensitivity                |
| <b>Consumer outcomes</b>       | Healthy meal uptake, taste satisfaction, affordability compliance, recommendation acceptance, repeat purchase | Measures user-facing success            |
| <b>Operational outcomes</b>    | Forecast accuracy, stockouts, wait time, capacity use, food waste                                             | Measures efficiency and feasibility     |
| <b>Financial outcomes</b>      | Margin, cost per meal, healthy meal revenue share, subsidy efficiency                                         | Measures economic viability             |
| <b>Sustainability outcomes</b> | Carbon footprint per meal, local/seasonal sourcing rate, total waste, packaging waste                         | Measures environmental performance      |
| <b>Supplier outcomes</b>       | Order variability, fill rate, lead-time reliability, spoilage                                                 | Measures upstream coordination          |
| <b>Governance outcomes</b>     | Fair access, stakeholder utility balance, explainability/trust                                                | Measures legitimacy and equity          |
| <b>Statistical analysis</b>    | Multiple replications; confidence intervals; pairwise tests and multi-scenario comparison                     | Tests significance                      |
| <b>Practical significance</b>  | % improvement, effect sizes, relative gain vs baseline                                                        | Tests real usefulness                   |
| <b>Trade-off analysis</b>      | Pareto frontier across health, taste, price, and waste                                                        | Tests multi-objective superiority       |
| <b>Robustness analysis</b>     | Stress tests under disruption, price shock, and demand shifts                                                 | Tests stability of improvements         |
| <b>Success criterion</b>       | MSRS outperforms baselines on key KPIs without major loss to any stakeholder group                            | Demonstrates balanced value co-creation |

### 8.3.1 Example hypotheses

- **H1:** The MSRS scenario yields significantly higher healthy meal uptake than the consumer-only recommender.
- **H2:** The MSRS scenario significantly reduces food waste and stockouts compared with rule-based menu planning.

- **H3:** The MSRS scenario improves the combined ecosystem performance score without reducing taste satisfaction or affordability.
- **H4:** The MSRS scenario produces more favorable Pareto trade-offs across health, taste, price, and sustainability than all baselines.

## 9 Illustrative Evaluation Results

### 9.1 Overview of comparative performance

To illustrate the managerial relevance of the proposed framework, we report a set of **hypothetical simulation results** comparing four scenarios: **no recommender**, **consumer-only recommender**, **rule-based healthy menu planning**, and the **full MSRS-enabled ecosystem**. Across repeated simulation runs, the MSRS scenario outperforms all baseline conditions on the majority of ecosystem-level indicators while avoiding major losses for any stakeholder group.

In the hypothetical results, the MSRS scenario increases **healthy meal uptake** by approximately **18–26%** relative to the no-recommender baseline and by **8–14%** relative to the consumer-only recommender. At the same time, **taste satisfaction** remains stable or improves slightly, suggesting that the system does not achieve health gains by sacrificing perceived meal quality. **Affordability compliance** also improves because the decision-support layer uses pricing, bundling, and substitution logic to keep recommended meals within student budget constraints.

### 9.2 Operational and sustainability results

Operationally, the MSRS scenario yields improvements in **forecast accuracy**, **stockout reduction**, and **food waste minimization**. In the hypothetical simulation, forecast accuracy improves from roughly **72–78%** in baseline settings to **84–89%** under the full ecosystem model. Plate and production waste fall by **12–21%**, primarily because the system aligns menu planning, demand forecasting, and procurement more effectively. Queue time remains within acceptable limits, indicating that healthier and more diversified offerings do not create excessive service bottlenecks.

The environmental results are also favorable. The MSRS scenario lowers the **carbon footprint per meal** through more targeted sourcing, better menu composition, and lower waste. In the hypothetical runs, carbon intensity declines by **9–16%**, while the share of **local/seasonal sourcing** rises under scenarios where sustainability receives moderate or high policy weight.

### 9.3 Stakeholder utility results

The simulated stakeholder-utility analysis shows that the full MSRS model produces the most balanced outcome across actors. Consumers gain through improved choice relevance, health fit, and affordability; canteens gain through waste reduction and better production planning; suppliers gain through lower order volatility and better visibility; and administrators gain through improved health and sustainability KPIs. Although some scenarios produce small trade-offs, the MSRS condition dominates baseline approaches on the combined ecosystem performance score and produces a more favorable Pareto frontier across **health**, **taste**, **price**, and **waste**.

### 9.4 Illustrative summary table

| Metric                              | No recommender | Consumer-only recommender | Rule-based planning | Full MSRS ecosystem |
|-------------------------------------|----------------|---------------------------|---------------------|---------------------|
| Healthy meal uptake (%)             | 31             | 39                        | 42                  | 49                  |
| Taste satisfaction (1–5)            | 3.9            | 4.0                       | 3.8                 | 4.1                 |
| Meals within budget (%)             | 68             | 73                        | 75                  | 82                  |
| Forecast accuracy (%)               | 74             | 77                        | 81                  | 87                  |
| Food waste per 100 meals (kg)       | 18.5           | 16.9                      | 15.8                | 13.9                |
| Carbon footprint per meal (kg CO2e) | 2.40           | 2.31                      | 2.20                | 2.02                |
| Supplier order variability index    | 1.00           | 0.94                      | 0.88                | 0.76                |
| Ecosystem performance score (0–100) | 56             | 64                        | 69                  | 81                  |

These results are illustrative and are intended to demonstrate the type of empirical pattern that would validate the framework once fully implemented and simulated.

## 10 Discussion and Engineering Management Implications

### 10.1 Engineering management implications

The hypothetical findings suggest that the proposed framework is highly relevant to engineering management because it shifts the design problem from isolated optimization to **ecosystem orchestration**. Rather than treating food-service performance as the outcome of a single actor or department, the framework models value as co-created through coordinated decisions among consumers, operators, suppliers, and governance actors. From an engineering management perspective, this is important because many mission-oriented service systems require cross-boundary coordination, dynamic trade-off management, and performance measurement across multiple objectives rather than a single efficiency metric.

### 10.2 Value co-creation as a design problem

The results also support the argument that **value co-creation** should be treated as an operationalizable design logic rather than merely a conceptual descriptor. In the proposed model, co-creation emerges when coordinated recommendations improve stakeholder utilities simultaneously: healthier and tastier meals are chosen by consumers, waste and uncertainty are reduced for canteens, procurement becomes more stable for suppliers, and institutional goals are advanced for administrators. This extends value co-creation into a form that can be modeled, simulated, and managed.

### 10.3 Why MSRS adds value beyond consumer personalization

A key implication of the comparative scenarios is that a **consumer-only recommender** is not sufficient for mission-oriented sustainable systems. Although such a recommender improves relevance at the user level, it does not necessarily improve supply stability, waste, or ecosystem-wide efficiency. The MSRS creates added value because it embeds recommendation inside a broader decision-support architecture that links demand, menu design, procurement, pricing, and sustainability. Thus, the main managerial contribution of the MSRS is not only better personalization, but better **coordination across stakeholder interfaces**.

### 10.4 Broader transferability

Although the present application focuses on campus food services, the framework is transferable to other service ecosystems in which multiple actors co-create value around mission-oriented outcomes. Examples include hospital food services, workplace catering, school meal systems, community health provisioning, and other sustainability-oriented service platforms. More broadly, the model offers an engineering management template for designing digital coordination

systems in settings where quality, affordability, operational efficiency, and sustainability must be balanced simultaneously.

## 10.5 Managerial implications

For engineering managers, the framework suggests that mission-oriented service innovation should be managed as an ecosystem orchestration problem rather than as an isolated product or operations problem. Managers should explicitly model stakeholder objectives, make exchange mechanisms visible, and use digital coordination tools such as MSRS to align design, sourcing, execution, and governance decisions. For operators, this means using recommendation intelligence not only to influence demand, but also to guide production scheduling, inventory planning, and service delivery. For regulators and institutional leaders, the framework highlights the importance of setting mission weights, fairness safeguards, and incentive mechanisms that shape ecosystem behavior without undermining stakeholder viability.

## 10.6 Governance implications

The manuscript also implies that governance is a core design variable in mission-oriented ecosystems. Recommendation systems operating across multiple actors require transparent stakeholder weighting, accountability for trade-offs, and explicit mechanisms for fairness, explainability, and consent. From a TEM perspective, this means the digital coordination layer should be designed not only for efficiency, but also for legitimacy and institutional trust.

## 10.7 Limitations

This study is primarily conceptual and simulation-oriented. The campus food-service setting serves as an illustrative application rather than a fully validated empirical case study. The hypothetical results are intended to demonstrate the type of system-level effects that the framework could generate under realistic assumptions, but they do not substitute for field implementation and empirical testing. In addition, stakeholder utility weights are likely to vary across service ecosystems and may be politically contested in practice. # Conclusion

This paper developed a value co-creation modeling framework for **mission-oriented sustainable service ecosystems** and illustrated its application to **healthy, tasty, and affordable campus food services**. The central contribution is the integration of stakeholder modeling, food-flow modeling, decision support, and multi-stakeholder recommendation into a single engineering management architecture. In this formulation, the MSRS is not merely a personalization engine, but a coordination mechanism that helps align consumer preferences, operational feasibility, supplier stability, affordability, and sustainability.

The paper contributes to engineering management by showing how mission-oriented ecosystems can be designed and governed as multi-stakeholder systems with measurable value co-creation

outcomes. It also demonstrates how recommendation systems can be reframed from user-centric tools into broader mechanisms for ecosystem governance and performance improvement. The proposed framework provides a structured basis for future empirical work, simulation experiments, and field implementation in campus food services and analogous service ecosystems.

## 11 Future Research Directions

Future research should extend and refine the present framework in five directions. First, the model should be implemented as a full **simulation environment or digital twin** in order to test sensitivity to demand volatility, policy weighting, supply disruptions, and consumer heterogeneity. Second, future studies should estimate stakeholder utility functions empirically using transaction data, survey data, interviews, and operational records. Third, field studies should examine whether the framework improves outcomes in real campus dining settings and whether observed gains are robust over time.

Fourth, future work should examine governance issues in greater depth, particularly fairness across user groups, explainability of recommendations, and the managerial legitimacy of stakeholder-weight assignments. Finally, the framework should be generalized beyond campus food services to other sustainable service ecosystems in which mission-oriented innovation depends on coordinated digital decision support. Together, these directions would move the framework from conceptual and simulation-based design toward validated engineering management practice.

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